

Executive Summary

In 2025, the European Union was one of the world regions hardest hit by cyberattacks. A variety of states such as (but not limited to) Russia, China, and Iran, as well as a broad spectrum of non-state actors continually engaged in operations in its cyberspace. This phenomenon, which continues to augment annually (ENISA, 2025), has set a clear tendency in cyberspace: threat actors are not only increasing, but also diversifying, collaborating, and introducing new attack models to persist in targeting European infrastructures.

Both cyberattacks and cyber espionage, which are used to conduct hybrid warfare, disinformation campaigns, data and financial theft, and to contribute to European societal destabilization have become a strategic geopolitical risk for the EU. These developments showcase that cyber threats target not just systems, but also societies; yet the EU's ability to convey the pertinence of cyberthreats to its citizens has remained deficient.

Ulrich Beck, in his work *Risk Society: Towards a New Modernity* (Beck, 1986), emphasizes the increasing reliance of modern societies on individuals to navigate complex and often invisible risks. Within this same framework, Herbert Simon highlights in *Models of Man* (Simon, 1957) that individuals are guided by the notion of bounded rationality, which compromises their ability to make optimal decisions derived from the information they process.

This tendency can clearly be observed in the wider European cybersecurity context: despite increasing and diversifying cyberthreats, limitations in digital and cyber literacy in the EU populations remain significant (European Commission, 2025).

To illustrate this, the European Union Agency for Cybersecurity has introduced regulations such as the NIS2 directive to enhance cyber practices in its critical sectors, as well as initiatives like the Digital Action Plan (2021–2027) and the European Cybersecurity Month to raise cybersecurity risk awareness among its citizens. Nevertheless, absence of basic cyber hygiene remains the most commonly exposed weakness by cybercriminals targeting the EU. This pertains to critical sectors like public administration, healthcare, finance, and transport, which represented 53% (ENISA, 2025) of EU cyberincidents in 2025 despite being covered by the NIS2 directive.

Similarly, social engineering attacks such as phishing still account for at least 60% of penetration vectors used in cyberattacks in the EU (ENISA, 2025). By the same token, (even advanced) attackers still exploit commonly used platforms (Signal, Telegram, and others) to perform malicious activities by using trivial points of entry. These vary from weak or reused passwords to absence of multi-factor authentication, to unrestricted VPNs, and others. (CYBERUK, 2025).

Absence of safe digital practices leading to cybercrime can be observed even in populations of the most cyber-resilient states in the EU. In Latvia, a country reputable for its ability to prepare for, respond to, and recover from cyber-attacks, for instance, financial theft through cybercrime in 2025 reached more than 17 million euros (CERT.LV, 2025) thanks to attackers exploiting basic vulnerabilities, a striking disproportion for a population of 1,8 million.

This illustrates how effectively cybercriminals can manipulate not only individuals' devices, but also their data, finances, behavior, and even emotions that allow for cybercrime to take place, confirming the belief that amateurs hack systems, but professionals hack people.

This persistence of rudimentary human vulnerabilities in the face of evolving multifaceted and hybrid threats aimed at both systems and societies exposes an important tension in the EU cyberspace and raises an important question: if cyber actors are becoming increasingly more persistent, diverse, and creative, why does cybersecurity awareness in the European Union remain stagnant?

This presentation will draw on elements and contradictions observed at the Computer Emergency Response Team of the Republic of Latvia (CERT.LV) that accentuate the pertinence and recurrence of human vulnerability in cyberspace. These will include references to monthly reports of cyberincidents in Latvia and the EU, communication work and public awareness campaigns targeting critical sector entities within the NIS2 directive framework, as well as simulation games for both students and professionals aimed at increasing cyberthreat awareness. These experiences combined ensure evidence-based analysis concerning the gap between expectations of European policy frameworks and actual user practices.

This presentation argues that European cybersecurity is systemically bounded by its dependence on individual user behavior. It is structured around three core arguments. First, it will explain how and through what paths human behavior still forms the principal vulnerability in cybersecurity. Second, it will examine the limits of cybersecurity awareness and education (both in Latvian and a wider EU context), highlighting the gap between formal expectations and reality. Third, it will underline how these vulnerabilities on micro level expand into meaningful geopolitical risks for the EU, despite governing frameworks such as the NIS2 directive, before finally addressing cross-cutting solutions to navigate the risks arising as a result.